Effects of grass–endophyte symbiosis and herbivory on population demography across climatic and geographic gradients

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Abstract

Interactions between plants and fungi are ubiquitous in nature and have significant effects on plant fitness. However, the extent to which variation in fungal symbiosis affects plant demography across environmental gradients remains understudied. Using three cool-season grass species that host seed-transmitted fungal endophytes, we combined geographically-distributed field experiments across an aridity gradient and Bayesian modeling of demographic performance to test whether plant-fungi interactions affect host demography and, consequently, determine geographic range limits. We found that all vital rates decreased toward the species' range edges, indicating reduced demographic performance in these regions. This decline was more pronounced in individuals hosting endophytic symbionts, suggesting that plant-fungi interactions that are beneficial under benign conditions may become parasitic under stressful conditions. Our results highlight that host resistance at range limits[1] may be hindered by the presence of a mutualist.

[1] *Tom wants us to think about this a bit more.*

Introduction

Plant-microbe symbioses are widespread and ecologically important. These interactions are famously context-dependent, where the direction and strength of the interaction outcome depends on the environment in which it occurs (Bronstein, 1994; Hoeksema and Bruna, 2015). Plant-microbe interactions that are beneficial under stressful conditions may become neutral or parasitic under benign conditions (Giauque et al., 2019). Under biotic stress (e.g., herbivory), endophyte symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing insect growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly under abiotic stress (e.g., drought), symbionts can increase their host tolerance (Clay and Schardl, 2002). For example, mycorrhizal fungi form extensive networks that increase root surface area, helping plants absorb more water and nutrients such as phosphorus and nitrogen (Amijee et al., 1989). Symbiotic microbes also stimulate plants to produce antioxidant enzymes like superoxide dismutase and catalase that reduce oxidative damage caused by stresses such as drought, salinity, and heat (Ruiz-lazona et al., 1996). All these mechanisms help the host species better buffer against environmental stochasticity (Fowler et al., 2023).

Context dependence raises the hypothesis that plant-microbe interactions are likely to vary across environmental gradients, from range core to range edge, with significant implications for host range expansion at leading edges or resilience at trailing edges. Because range-edge environments are often more stressful due to factors such as extreme temperatures and limited soil nutrients (Hampe and Petit, 2005), the role of microbial symbionts may be especially important in these zones. If microbial symbiosis provides greater benefits under environmental stress, then symbionts could enhance the suitability of range-edge environments, potentially extending host range limits (Allsup et al., 2023; Rudgers et al., 2020). For instance, fungal endophytes improve the survival of *Bromus laevipes* populations in dry conditions, increasing their drought resistance at the range edge and thereby extending the species' geographic range (Afkhami et al., 2014; David et al., 2019). Even if the symbiont does not directly improve host survival, it may still enhance host population growth over time by increasing relative growth and reproduction, potentially offsetting the negative effects of lower survival rates (Yule et al., 2013). Conversely, if microbial symbiosis is costly for the host at range edge, symbionts could constrain host range (Bennett and Groten, 2022; Benning and Moeller, 2021a,b). For instance, symbiotic microbes, especially mycorrhizal fungi, require carbon subsidies from the plant (up to 20% of photosynthetically fixed carbon) (Soudzilovskaia et al., 2015; Thirkell et al., 2020). At range edges, photosynthesis may be limited, making the carbon cost harder to bear. Under such conditions, plants may reduce symbiosis, leading to host poor performance.

Despite growing interest in the role of symbiosis in host performance, few studies examined in natural conditions how abiotic and biotic stressors shape the effects of symbionts on host demography across geographic ranges (Louthan et al., 2015). Moreover, it remains unclear which stressor is most critical in eliciting the benefits of endophyte symbiosis, or how multiple stressors may interact to influence host demography. Herbivory is a pervasive biotic factor that can significantly affect host performance (Benning et al., 2019) and mediate the outcomes of symbiotic interactions across species' ranges. Herbivores reduce host fitness through tissue damage, defoliation, and altered resource allocation (Maron and Crone, 2006), but they may also shape the selective landscape in ways that favor symbiont persistence or transmission. For example, herbivory can enhance the fitness benefits of endophyte infection by triggering host defense pathways or promoting vertical transmission of the symbiont (Agrawal et al., 1999; Gundel et al., 2020). These effects may be especially pronounced at range margins, where plant populations often encounter intensified biotic pressures. If the ecological context of herbivory alters the net fitness effects of symbiosis, then interactions between herbivory and abiotic stress (e.g., drought) may play a critical role in determining population persistence and thereby species' range limits.

Working across a precipitation gradient in the south-central US, we investigated how endophyte symbiosis affects their host demography from core to edge. To do so, we studied the symbiotic association between three native cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus* and *Poa autumnalis*) and their vertically transmitted fungal symbionts (*Epichloë* spp.). Our experiment was design to address the following:

1. Do plants with endophytes experience less damage from biotic stressors and maintain higher vital rates compared to plants without endophytes, when biotic stressors are present?
2. Under combined biotic and abiotic stress, do the mutualistic benefits of endophytes give host plants a performance advantage compared to those without endophytes?
3. Under combined biotic and abiotic stress, do the added physiological demands and resource costs of maintaining endophyte associations reduce overall plant performance, resulting in endophyte-infected plants performing worse than uninfected plants?

Materials and methods

Study system

Our study was conducted across seven sites along a geographic and climatic gradient spanning Louisiana and Texas in the United States (Fig. 1; Fig.S-1). At each site, we established eight plots (1.5 m $\times$ 1.5 m), with each plot containing 18 individuals of a single species. We used three cool-season perennial grasses native to woodland and prairie habitats of eastern North America (Shaw, 2011): *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (subfamily Pooideae). They typically host seed-transmitted fungal endophytes. In summer–fall 2021, seeds were collected from naturally symbiotic populations (source populations) of the three focal host species (Fig. 1). These seeds were either heat-treated to eliminate symbionts or left untreated to generate symbiont-free (E^{-}) and symbiotic (E^{+}) plants from the same genetic lineages as the source populations.

Across all sites, each plot was randomly assigned one of four initial endophyte frequency treatments (80%, 60%, 40%, or 20%) and one of two herbivory treatments (herbivore exclusion or herbivore access). We ensured that all plots received comparable quantities of source plant material. This experimental design was informed by natural patterns of endophyte prevalence observed in Texas, where the proportion of symbiotic individuals has been reported as 86.55% in *A. hyemalis* (ranging from 77.16% to 93.5%) (Donald et al., 2021), 53% in *E. virginicus* (ranging from 10% to 100%) (Sneck et al., 2017), and 96% in *P. autumnalis* (Rudgers et al., 2009). Additionally, fungal genotyping has confirmed the presence of biosynthetic pathways for secondary metabolites such as peramine, loline, and ergot alkaloids, which may enhance host resistance to drought and herbivory (Beaudry, 1951)[2]. Full details of the study design are provided in the supplementary materials (Supplementary Method S.1).

Demographic data collection

We collected demographic data including survival, growth, and reproduction in May and June of 2023, 2024 and 2025, which coincided with the flowering season of *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*. For each individual, plant survival was recorded as a binary variable (alive or dead), and plant size was measured as the number of living tillers. We recorded the number of inflorescences per plant and counted the number of spikelets on up to three inflorescences from three reproducing plants. Spikelet counts were limited to three reproducing

[2] *I am trying to justify why we choose the starting endophyte prevalence*

tillers per plot due to the time-consuming nature of this measurement. We used the number of spikelets from these three tillers to estimate the average number of spikelets per plant.

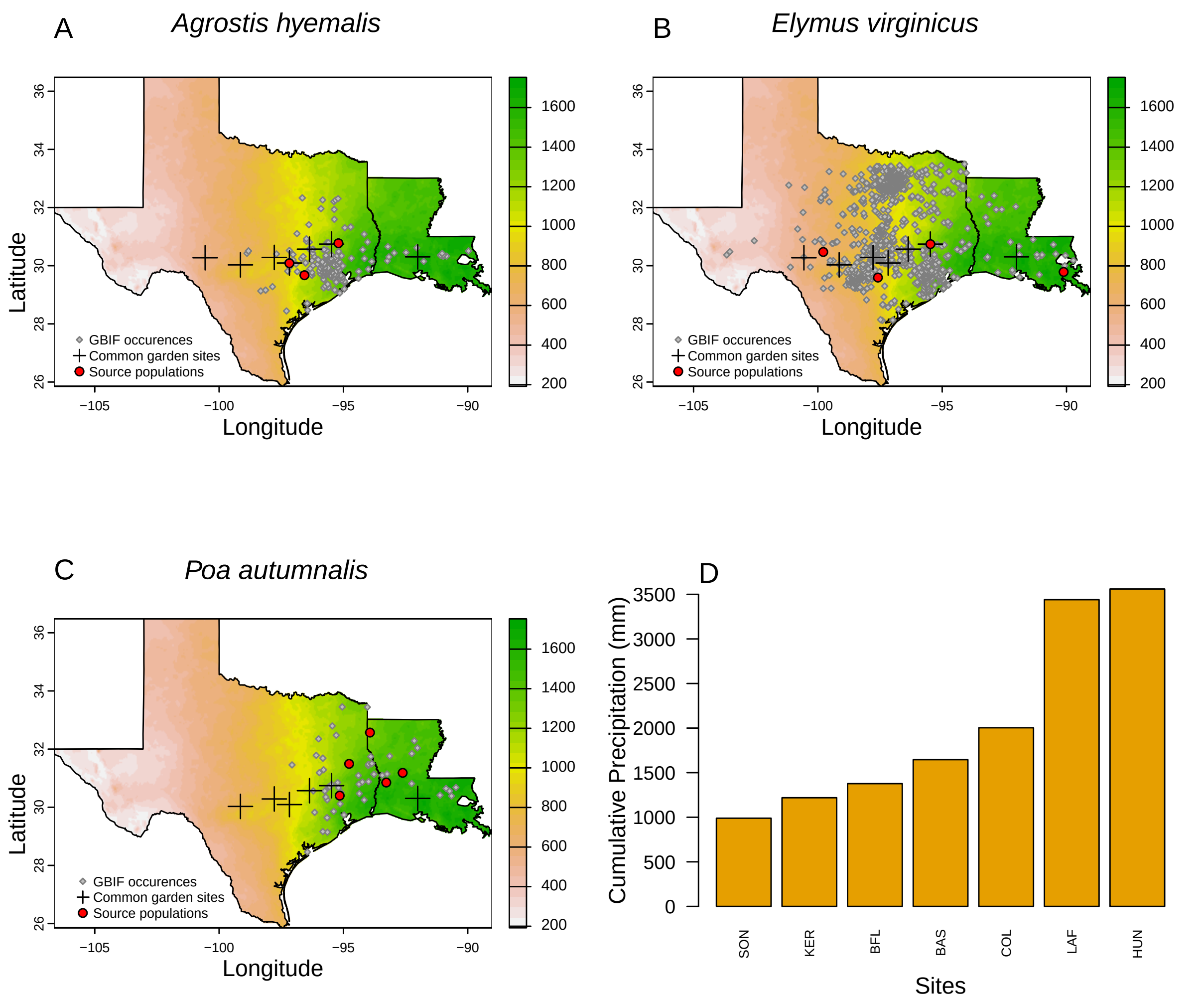


Figure 1: Distribution of common garden sites along the longitudinal aridity gradient in the central and southern Great Plains for: A) *Agrostis hyemalis* (AGHY), B) *Elymus virginicus* (ELVI), and C) *Poa autumnalis* (POAU). Red dots indicate the locations of source populations, while grey dots represent the Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area. D) Relationship between the distance from the range center and longitude for each species.

Models building and models selection

To assess how stress associated with aridity and herbivory affects plant–fungal interactions, we developed four candidate models for each vital rate (survival, growth, and fertility). Each vital rate was modeled using a grand mean intercept (β_0), slopes for variation in each covariate (β_1,

β_2 and β_3), and interaction terms between covariates (β_4, β_5 and β_6). Each model included normally distributed random effects to account for site-to-site variation ($\phi \sim N(0,\ \sigma_{site})$), plot-to-plot variation ($\rho \sim N(0,\ \sigma_{plot})$), and source-to-source variation associated with the provenance of transplants used in the common garden ($\omega \sim N(0, \sigma_{source})$) (Eq. 1).

$$
\begin{aligned}
\text{Model 1:}\quad & \mu = \beta_0^{\text{sp}} + \beta_{\text{P}}^{\text{sp}} P + \beta_{\text{endo}}^{\text{sp}} \text{endo} + \beta_{\text{herb}}^{\text{sp}} \text{herb} \\
& + \beta_{\text{endo}\times\text{P}}^{\text{sp}} \text{endo} \cdot P + \beta_{\text{herb}\times\text{endo}}^{\text{sp}} \text{herb} \cdot \text{endo} + \beta_{\text{herb}\times\text{P}}^{\text{sp}} \text{herb} \cdot P \\
& + \beta_{\text{P}^2}^{\text{sp}} P^2 + \beta_{\text{endo}\times\text{P}^2}^{\text{sp}} \text{endo} \cdot P^2 + \phi^{\text{sp}} + \omega^{\text{sp}} + \rho^{\text{sp}} \\
\text{Model 2:}\quad & \mu = \beta_0^{\text{sp}} + \beta_{\text{GD}}^{\text{sp}} \text{GD} + \beta_{\text{endo}}^{\text{sp}} \text{endo} + \beta_{\text{herb}}^{\text{sp}} \text{herb} \\
& + \beta_{\text{endo}\times\text{GD}}^{\text{sp}} \text{endo} \cdot \text{GD} + \beta_{\text{herb}\times\text{endo}}^{\text{sp}} \text{herb} \cdot \text{endo} + \beta_{\text{herb}\times\text{GD}}^{\text{sp}} \text{herb} \cdot \text{GD} \\
& + \phi^{\text{sp}} + \omega^{\text{sp}} + \rho^{\text{sp}} \\
\text{Model 3:}\quad & \mu = \beta_0^{\text{sp}} + \beta_{\text{MD}}^{\text{sp}} \text{MD} + \beta_{\text{endo}}^{\text{sp}} \text{endo} + \beta_{\text{herb}}^{\text{sp}} \text{herb} \\
& + \beta_{\text{endo}\times\text{MD}}^{\text{sp}} \text{endo} \cdot \text{MD} + \beta_{\text{herb}\times\text{endo}}^{\text{sp}} \text{herb} \cdot \text{endo} + \beta_{\text{herb}\times\text{MD}}^{\text{sp}} \text{herb} \cdot \text{MD} \\
& + \phi^{\text{sp}} + \omega^{\text{sp}} + \rho^{\text{sp}}
\end{aligned}
\tag{1}
$$

where $sp \in \{1,2,3\}$ represents the species index, P denotes Precipitation, GD refers to distance from geographic center, and MD stands for mahalanobis distance, endo indicates Endophyte status (absence or present), and herb represents Herbivory (fenced and unfenced). The terms $\phi^{(sp)}$, $\omega^{(sp)}$, and $\rho^{(sp)}$ are plot-level, population-level, and site-level random effects, respectively, for species sp.

Survival was modeled using a Bernoulli distribution, growth with a Gaussian distribution, flowering with a negative binomial distribution, and fertility (number of spikelets) also with a negative binomial distribution. To assess the goodness-of-fit of the models, we performed posterior predictive checks (Berkhof et al., 2000; Gelman et al., 2000). All models effectively captured key aspects of the data, including means, standard deviations, and quantiles (Fig. **??**, Fig. **??**, Fig. **??**, Fig. **??**). To determine the best model for each vital rate, we compared the four models using leave-one-out cross-validation (LOOCV) (Vehtari et al., 2017). LOOCV combines validation and training by using one observation for testing and the remaining $n-1$ observations for training. This process is repeated n times once for each observation resulting in n fitted models (Silva and Zanella, 2024). The test error estimate from LOOCV is obtained by averaging the prediction errors across all n models (Eq. 2).

$$CV_n = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

All models were implemented in R (R Core Team, 2023) using Stan via the RStan interface (Stan Development Team, 2024).

Results

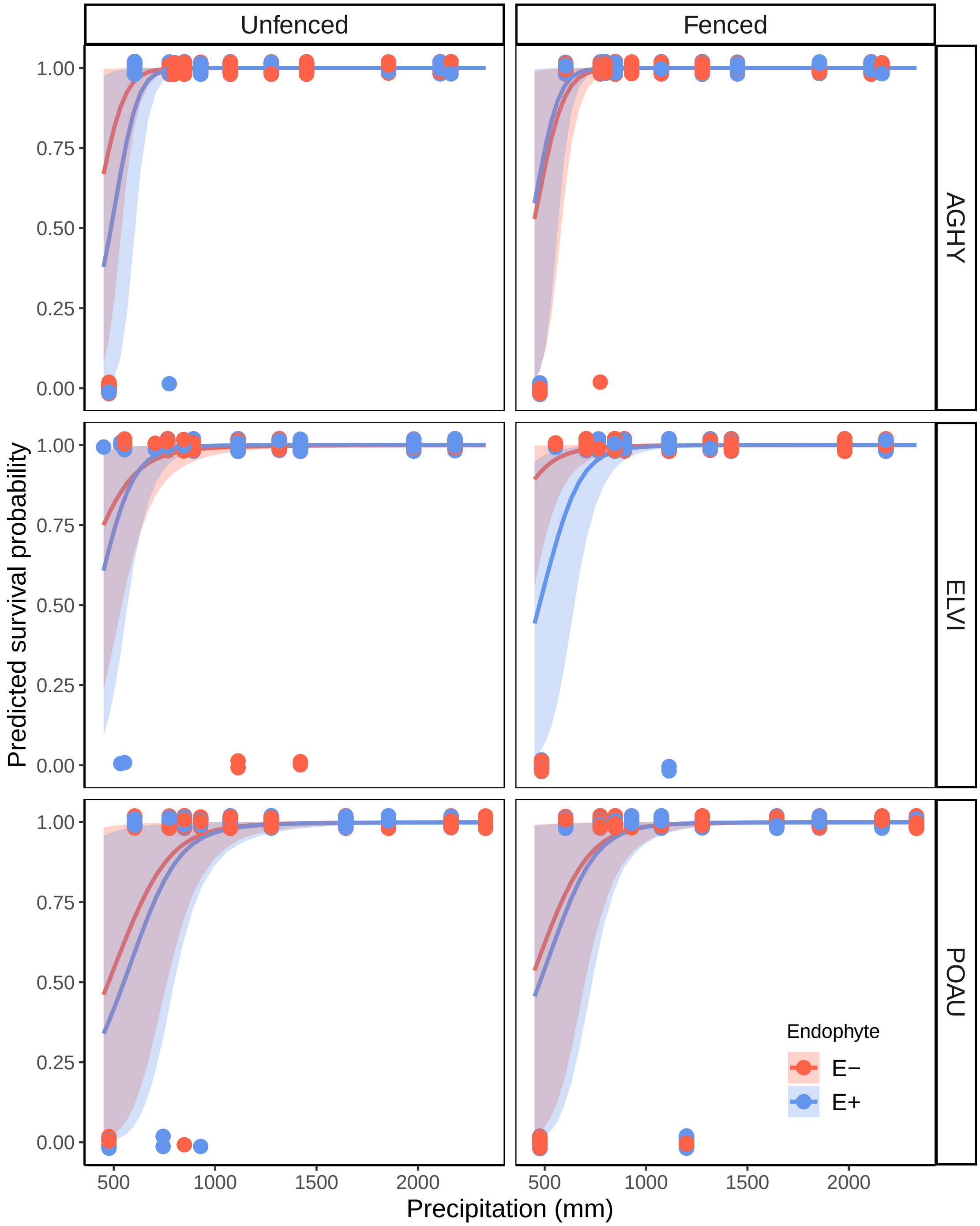
We tested the effects of precipitation, endophyte association (E^- vs. E^+), and herbivore exclusion (fenced vs. unfenced) on survival, growth, and reproductive traits in three cool-season grass species: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*. Our findings indicate that the influence of endophytes is context-dependent, varying from beneficial to neutral or even detrimental depending on species identity and herbivore presence.

At low precipitation levels representing species range edge, E^- individuals exhibited higher survival rates than E^+ individuals (Fig. 2), suggesting a cost of endophyte association under drought stress. This cost was partially offset in unfenced plots, where E^+ individuals showed comparable or greater survival relative to E^- individuals, potentially reflecting endophyte-conferred resistance to herbivory. Across all species, survival was generally higher in fenced plots, underscoring the negative impact of herbivores on plant persistence.

Patterns of growth were consistent across species and treatments. Across the full precipitation gradient, E^+ individuals had higher relative growth than E^- individuals (Fig. 3), suggesting a general growth advantage associated with endophyte presence. Moreover, plants in unfenced plots exhibited higher annual growth rates compared to those in fenced plots, possibly due to compensatory growth following herbivory or altered resource allocation strategies.

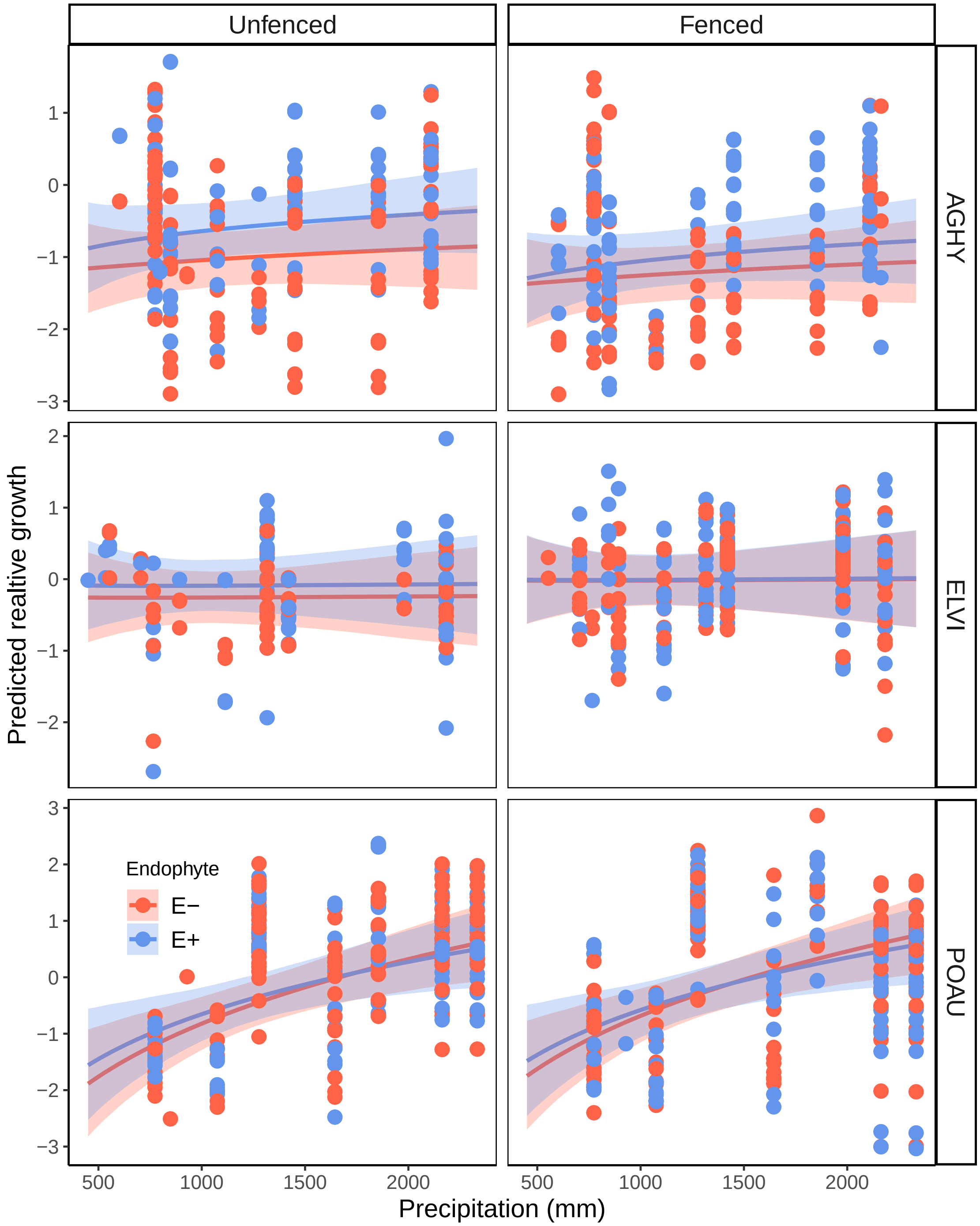
Inflorescence production tended to be higher in E^+ individuals than in E^- individuals under wet conditions, but this advantage declined under drier conditions (Fig. 4). This pattern suggests that the reproductive benefit of endophyte association is contingent on water availability. Fencing had a modest overall effect on flowering, although *Poa autumnalis* produced more inflorescences in fenced plots.

Spikelet production revealed a cost of endophyte association under dry conditions, particularly in *Elymus virginicus*, where E^+ individuals produced fewer spikelets than E^- individuals (Fig. 5). Additionally, fenced plots supported higher spikelet production across all species, further supporting the conclusion that herbivory suppresses reproductive output.



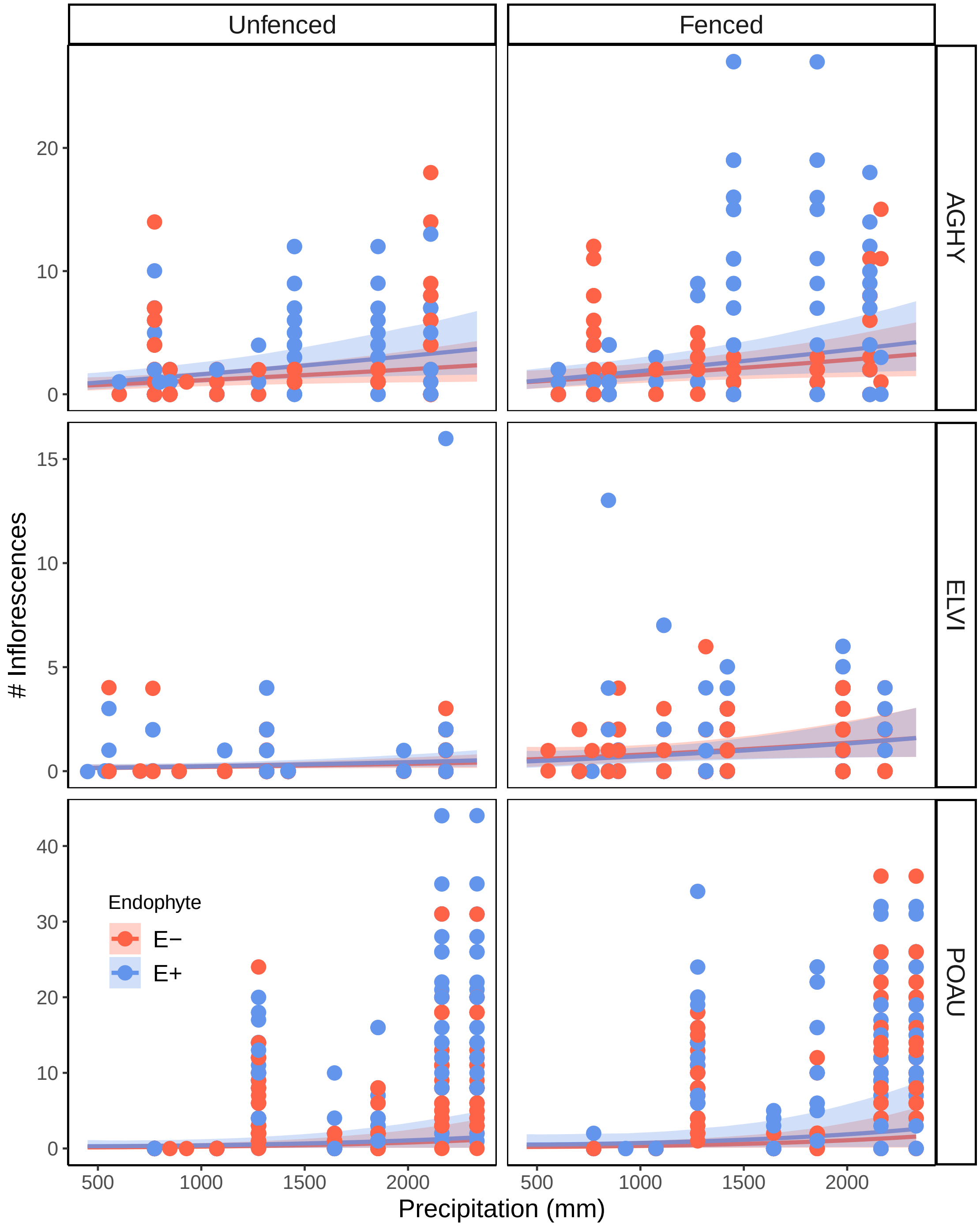
Unfenced
Fenced
AGHY
ELVI
POAU
Predicted survival probability
1.00
0.75
0.50
0.25
0.00
500
1000
1500
2000
Precipitation (mm)
Endophyte
E−
E+

Figure 2: XXX



Unfenced
Fenced
AGHY
ELVI
POAU
Predicted realtive growth
Precipitation (mm)
Endophyte
E−
E+
1
0
−1
−2
−3
2
1
0
−1
−2
3
2
1
0
−1
−2
−3
500
1000
1500
2000
500
1000
1500
2000

Figure 3: XXX



Unfenced
Fenced
AGHY
ELVI
POAU
Inflorescences
Precipitation (mm)
Endophyte
E−
E+

Figure 4: XXX

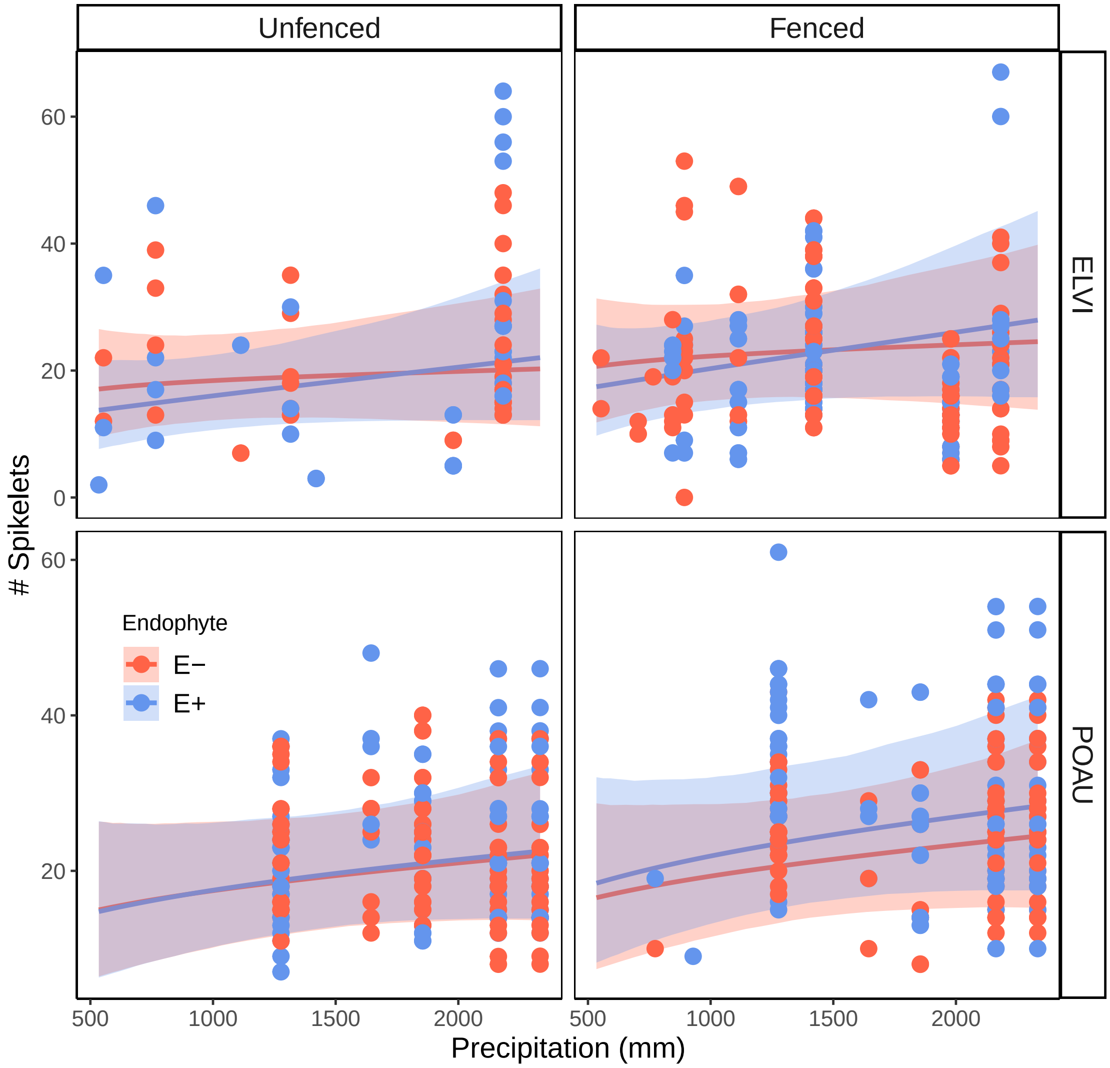


Figure 5: XX

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Supporting Information

S.1 Supporting methods

Study design

Experimental Design To understand the demographic effects of endophyte symbiosis from core to edge of the host range, we established common gardens at 7 sites across the geographic range of *Elymus virginicus* (Fig. S-1). Experimental sites spanned an aridity gradient (temperature gradient). Common gardens were established in 8 plots per site. Plots were 1.5m * 1.5m and the area was tilled of existing vegetation to control for native plant competition. Plots were also selected in shaded areas under tree canopy or near shrubs to mimic the natural environmental of the species. In each plot, we planted 15 individuals of *E. virginicus* approximately 15 cm deep in an evenly spaced 4*4 grid pattern, with positions randomly assigned. For each plot, we randomly assigned a starting endophyte frequency (80%, 60%, 40%, 20%)[3] and herbivory treatment (herbivores exclusion and herbivores accessibility). We ensured that all plots had comparable quantities of source populations. After establishing the plots, we watered the plants and recorded initial tiller counts, flowering status and plot position, endophyte status, source population of each individual plant. For herbivory exclusion plots, we enclosed them with 1.2m tall mesh fencing to prevent browsing by vertebrate herbivores and sprayed the plots with insecticide. For herbivores accessibility plots (control treatment), we half enclosed the plots with the mesh netting.

Source populations and Identification of individual endophyte status Plants used in the common garden experiment were derived from natural populations throughout the native range in the south-central US. At each of these natural populations we collected seeds. Some of the seeds of *E. virginicus* were heat treated to produce endophyte negative plants (E^-) . To do so, we placed these seeds in a drying oven set at 60°C for approximately five days (120 hours). While this method eliminates the endophytes from all individuals, it does not affect seed viability. All seeds (both heat-treated and non-heat-treated) were planted in the Rice University greenhouse. Seedlings were regularly fertilized every two weeks. The seedlings were then vegetatively propagated to produce enough individuals for your experiment (N = 840). Before planting in the field, we confirmed the endophyte status (E^+ or E^-) of all seedlings using either microscopy or an immunoblot assay. This was necessary due to the varying success of the heat treatment and differences in the prevalence of endophytes between

[3] *Do we need a schematic of one replicate of the experimental design?*

the natural populations. Leaf tissues were stained with aniline blue lactic acid and viewed under a compound microscope at 200x-400x to identify fungal hyphae. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of Epichloë spp. and chromagen to visually indicate presence or absence. Both methods yield similar detection rates.

S.2 Supporting Tables

Table S-1: Populations and Coordinates

Population	Location	Coordinates	NE+ / NE-
Sam Houston State University Field Station	Huntsville, TX	30°44’30.5”N, 95°28’28.2”W	NE+ = 39, NE- = 50
Texas Tech University	Junction, TX	30°28’18.2”N, 99°47’01.7”W	NE+ = 49, NE- = 83
Palmetto State Park	Palmetto, TX	29.591623, -97.584781	NE+ = 250, NE- = 192
Jean Lafitte National Historical Park and Preserve	Jean Lafitte, LA	29.785105, -90.114933	NE+ = 83, NE- = 94

Table S-2: Model comparison by vital rate using ELPD (Expected Log Pointwise Density) differences. The values in the **elpd_diff** column represent the difference in the model's ELPD relative to the best model for each vital rate, with positive values indicating a worse model fit. The **se_diff** column shows the standard error of the ELPD difference. Best models for each vital rate are bolded. Model selection is based on the ELPD difference and associated standard error, with smaller ELPD differences and lower standard errors indicating better model fit.

Vital rate	Model	elpd_diff	se_diff
Survival	**model1**	**0.0**	**0.0**
	model2	-1.0	1.5
	model3	-1.0	1.5
Growth	**model3**	**0.0**	**0.0**
	model2	-1.3	2.0
	model1	-2.1	2.2
Flowering	**model2**	**0.0**	**0.0**
	model3	0.0	3.2
	model1	-1.3	3.9

S.3 Supporting Figures

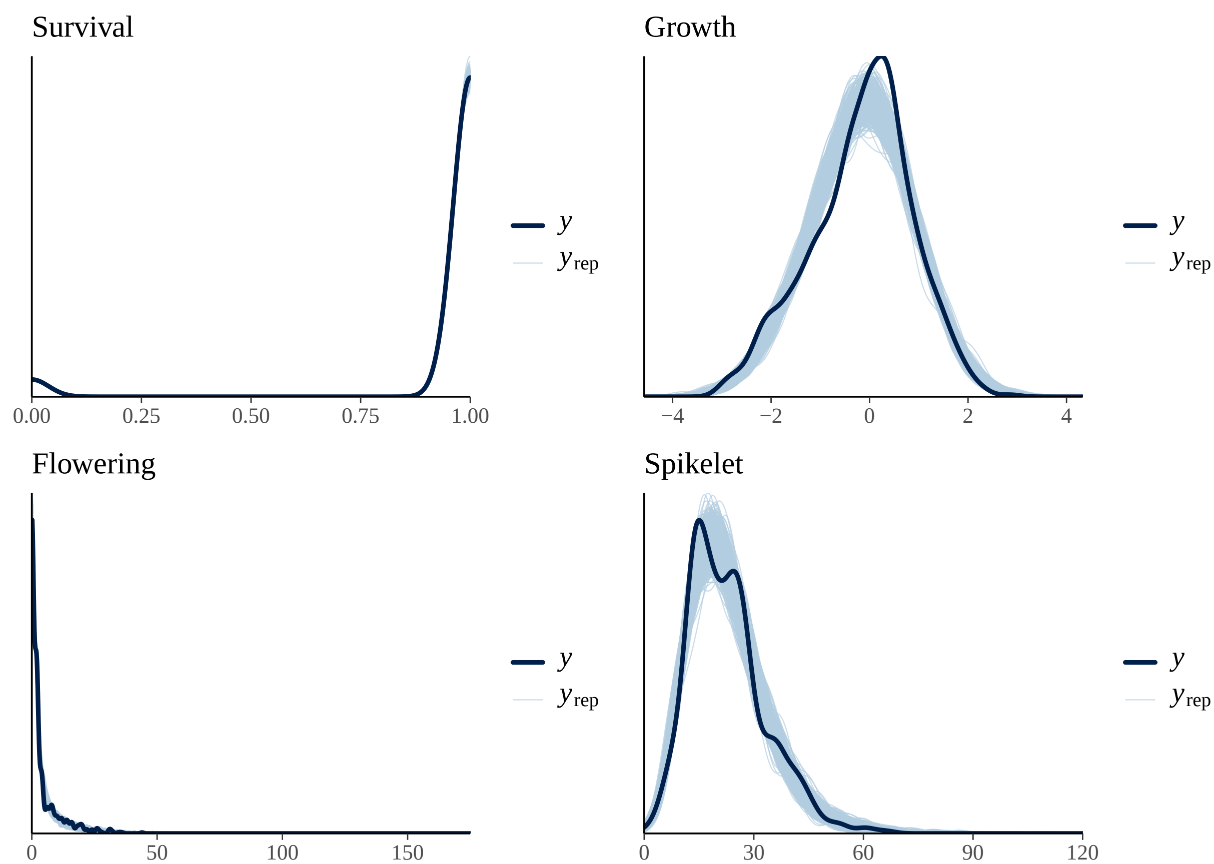

Figure S-1: (A-C) Relationship between distance from geographic center and longitude for the focal species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*). A generalized additive model (GAM) with a smooth term was used to capture the non-linear trends in these relationships for each species. The solid lines represent the fitted GAM with shaded regions showing the 95% confidence interval. D) displays the relationship between the distance from geographic center and the distance from niche centroid for all species also fitted with a GAM on a log scale. The p-value for the smooth term in the GAM model is p=0.0166**, indicating a significant relationship between log-transformed geographic distance and Mahalanobis distance.

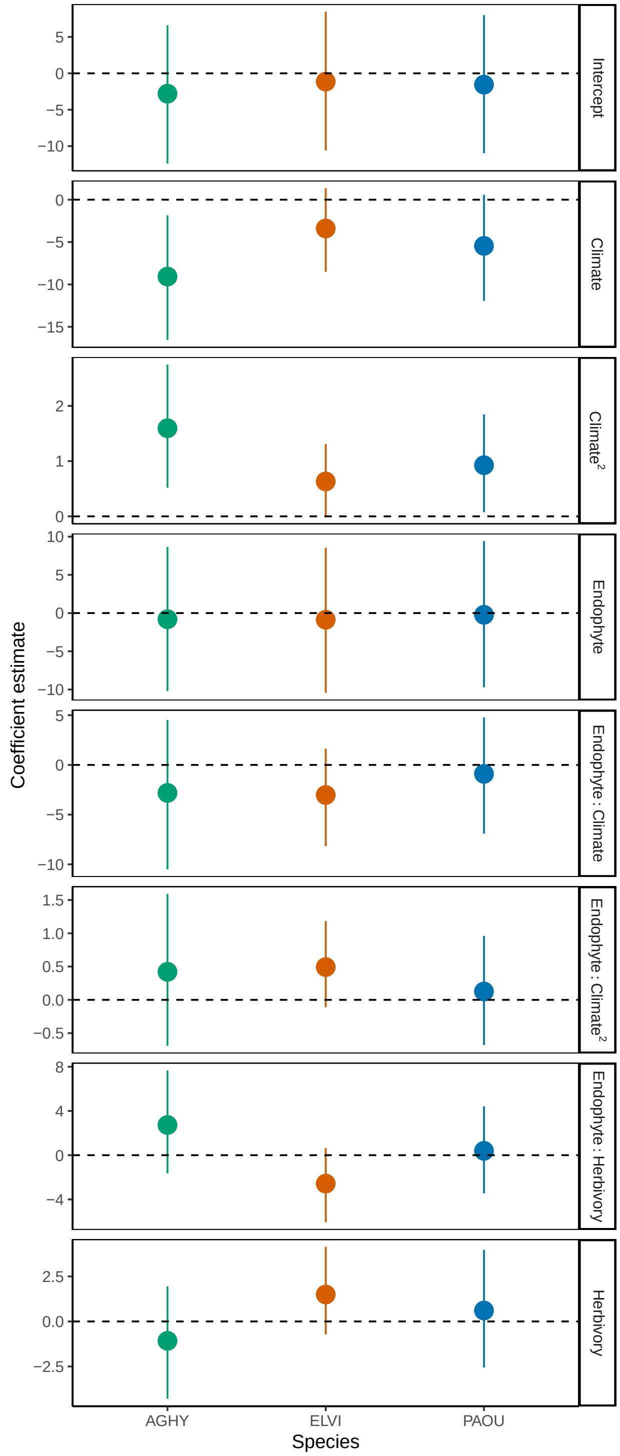

Figure S-2: Coefficient estimates from Bayesian models linking survival and geographic distance. Points represent posterior means and lines indicate 95% credible intervals. Species abbreviations: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (PAOU). Horizontal dashed lines denote zero effect.

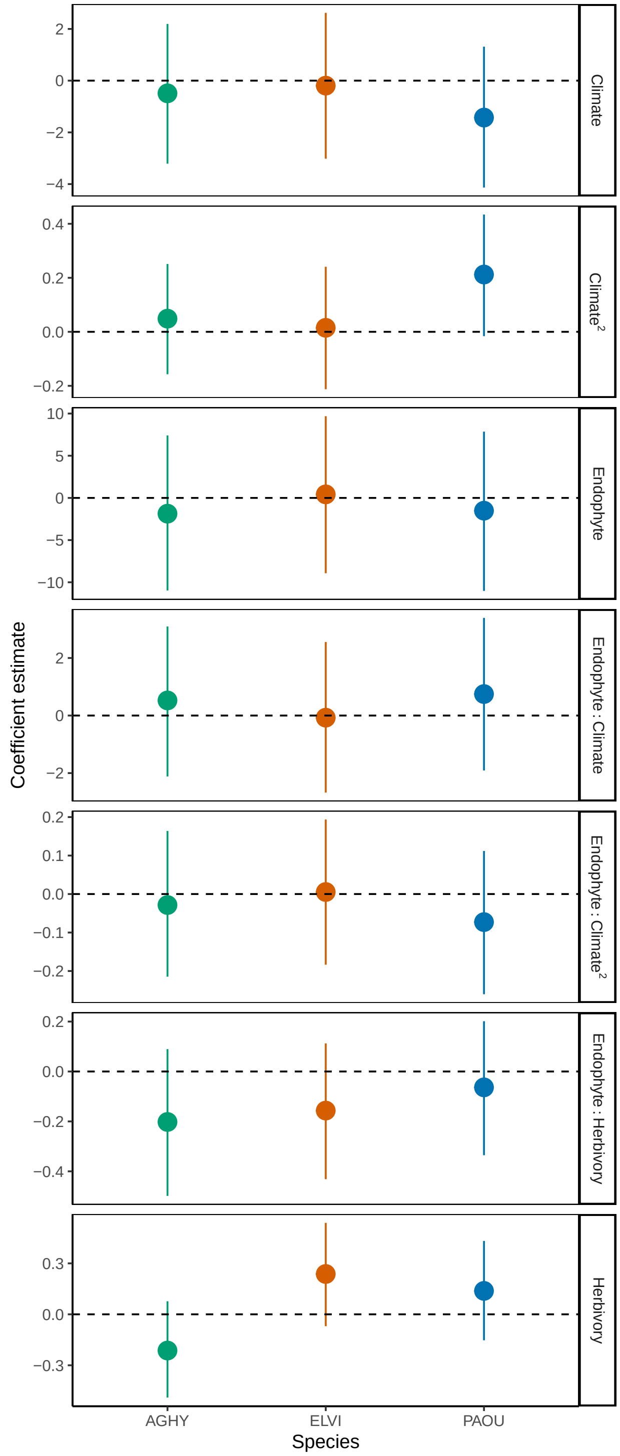

Figure S-3: Coefficient estimates from Bayesian models linking growth and geographic distance. Points represent posterior means and lines indicate 95% credible intervals. Species abbreviations: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (PAOU). Horizontal dashed lines denote zero effect.

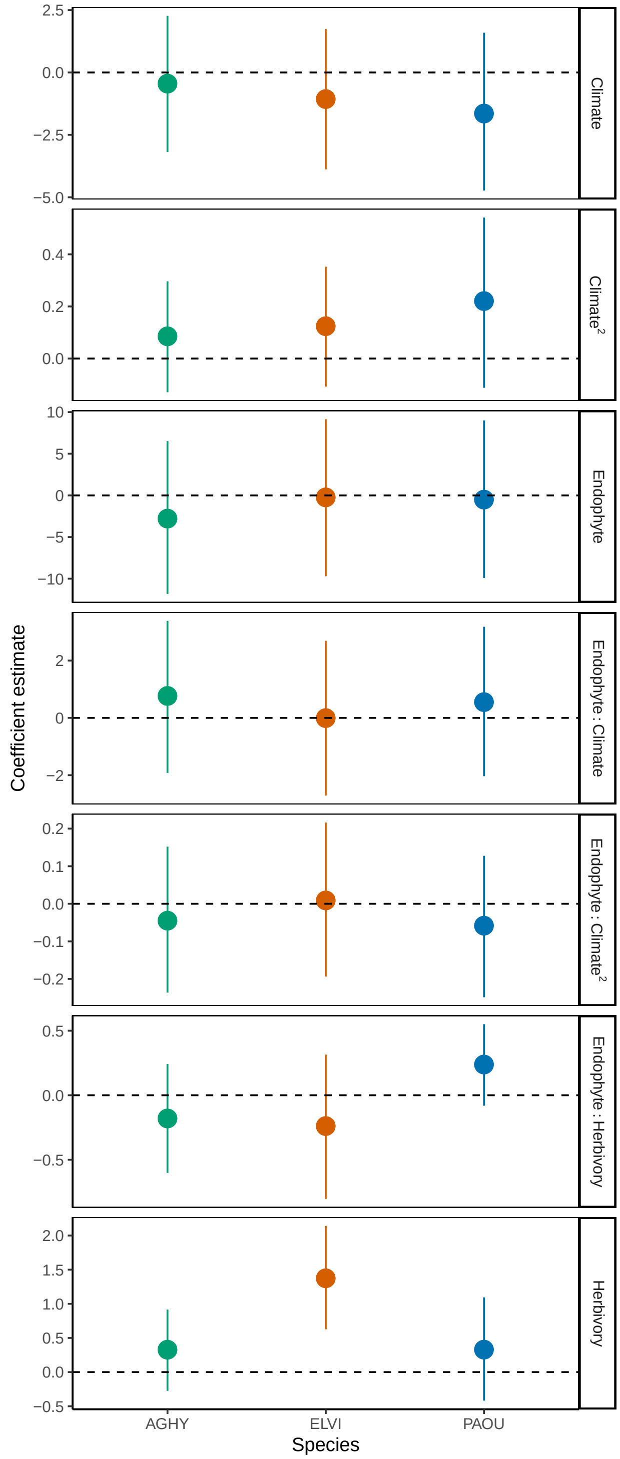

Figure S-4: Coefficient estimates from Bayesian models linking flowering and geographic distance. Points represent posterior means and lines indicate 95% credible intervals. Species abbreviations: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (PAOU). Horizontal dashed lines denote zero effect.

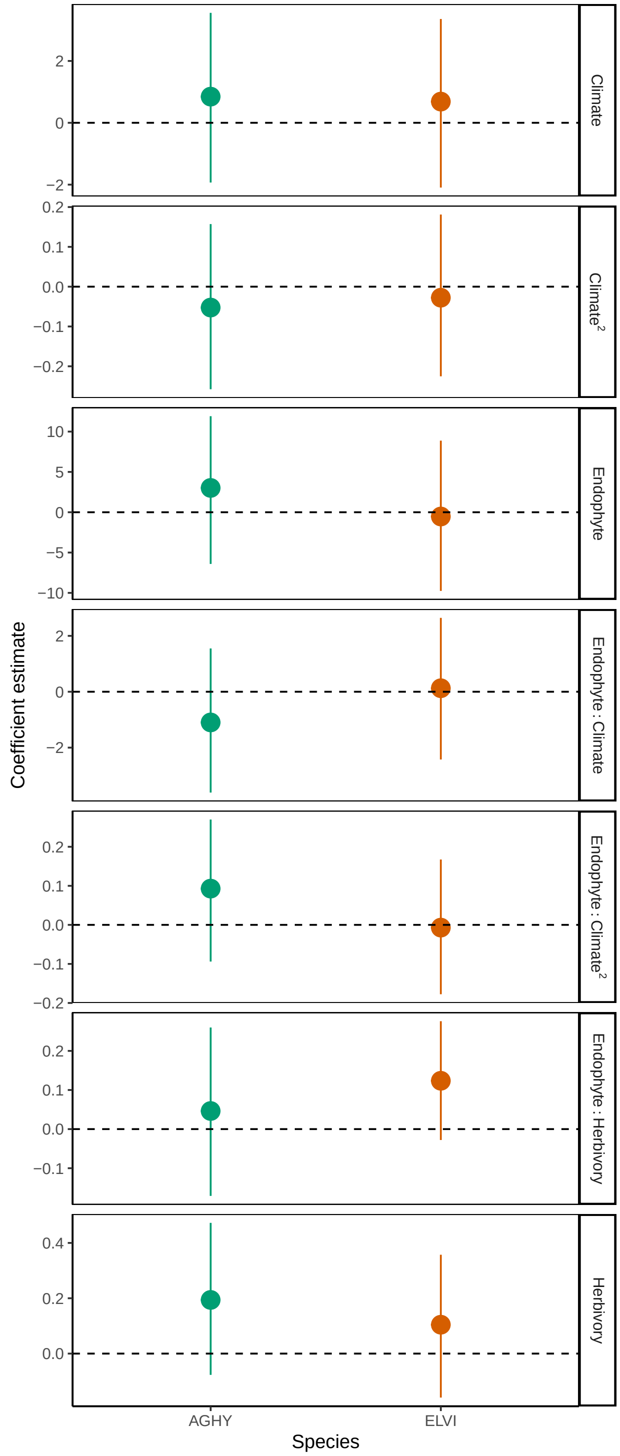

Figure S-5: Coefficient estimates from Bayesian models linking flowering and geographic distance. Points represent posterior means and lines indicate 95% credible intervals. Species abbreviations: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (PAOU). Horizontal dashed lines denote zero effect.